

Code Guru

Team Setup Guide

Everything you need to run PairPath and Study Guider on your own machine - with or without Cloudflare.

1. What you are setting up

Code Guru is four services in separate repositories: **Code Coach**, **Study Guider**, **PairPath** and the **Adaptive Gamification Engine**. All four are integrated against the same login.

The one thing to understand before anything else: **Code Coach owns login for the entire platform**. It is the only service that checks passwords. Study Guider and PairPath do not have their own accounts - they receive a token that Code Coach issued and ask Code Coach whether it is valid. There is a single web login screen, the **CodeGuru Portal**, which lives in the code-coach repo.

This means **you must run Code Coach locally**, even if you are only working on Study Guider or PairPath. Without it, every authenticated route in your service returns 503. The good news is that Code Coach needs no configuration at all to run locally - see section 5.

Ports

Service	Port	Who needs it
Code Coach backend	8000	Everyone. The identity provider.
CodeGuru Portal (login UI)	4200	Everyone, unless you use your own dev-login page.
Study Guider backend	8010	Study Guider devs.
Study Guider frontend	5173	Study Guider devs.
PairPath API	3001	PairPath devs.
PairPath frontend	3000	PairPath devs.
PairPath ml-service	8020	PairPath devs, only for live pair sessions.
Gamification API	3002	Gamification devs.
Gamification frontend	5174	Gamification devs.
Gamification ml-service	8030	Gamification devs, for difficulty prediction.

Port 8000 is taken by Code Coach

PairPath's ml-service README says to run it on port 8000. Do not. That is Code Coach's port. Start the ml-service on **8020** and make sure `ML_SERVICE_URL` in `Pair_Path/api/.env` points at 8020 - its built-in default is 8000, which would send ML requests to Code Coach.

2. Before you start

Check you have these. Versions below are what the team is using.

Tool	Check with	Needed for
Node.js 20+	<code>node --version</code>	All frontends, PairPath API, the portal

Tool	Check with	Needed for
Python 3.12+	<code>python --version</code>	Code Coach, Study Guider, ml-service
Git	<code>git --version</code>	Everything
Java 17+ (optional)	<code>java -version</code>	Only to run Code Coach's VS Code extension

3. Get the code

Clone all three repositories side by side, into one parent folder. The folder layout matters for nothing technical, but it keeps everyone's paths comparable.

```
mkdir Code_Guru && cd Code_Guru

git clone https://github.com/R26-SE-036/code-coach.git
git clone https://github.com/R26-SE-036/Study-Guider.git
git clone https://github.com/R26-SE-036/Pair_Path.git
git clone https://github.com/R26-SE-036/adaptive-gamification-engine.git
```

Every repo does its work on the **dev** branch. Never commit to main.

```
cd code-coach && git checkout dev && git pull && cd ..
cd Study-Guider && git checkout dev && git pull && cd ..
cd Pair_Path && git checkout dev && git pull && cd ..
cd adaptive-gamification-engine && git checkout dev && git pull && cd ..
```

Pulling later

Code Coach changes affect everyone, because it owns the login and the API contract. Run `git pull` in **code-coach** before you start work each day, not just in your own repo.

4. Two ways to run it

Pick one. Most of the time you want Mode A.

	Mode A - everything on your machine	Mode B - shared Code Coach via Cloudflare
When	Daily development. Building your own features.	Testing the real integration together, or demoing.
Code Coach	You run your own on localhost:8000.	One person runs it and shares a tunnel URL.
Accounts	Yours only. Disappear when you restart it.	Shared. Everyone sees the same students and triggers.
Setup	No credentials, no accounts, no tunnel.	Tunnel owner needs the Firebase key; you need their URL.
Cloudflare	Not needed at all.	Required.

In Mode A, Code Coach falls back to **in-memory storage** when it finds no Firebase credentials. That is exactly what you want while building: no setup, no shared state to corrupt, and a clean slate on every restart. It prints `Storage backend: in-memory` at startup so you always know which mode you are in.

What Mode A costs you

Because storage is in-memory, every account and every remediation trigger is lost when you restart Code Coach. Re-run the seed tool (section 9) - it takes about ten seconds.

5. Code Coach - everyone needs this

Backend

```
cd code-coach/backend

python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt

.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8000
```

No .env file is required for Mode A. Leave it absent and Code Coach uses in-memory storage. Confirm the startup line says Storage backend: in-memory. Then check `http://127.0.0.1:8000/health` returns `{"status": "ok"}`.

The portal (the login screen)

```
cd code-coach/portal

npm install
copy .env.example .env          # macOS/Linux: cp .env.example .env

npm run dev                     # http://localhost:4200
```

File to edit: code-coach/portal/.env

Variable	Mode A (local)	Mode B (Cloudflare)
VITE_CODE_COACH_URL	<code>http://127.0.0.1:8000</code>	the shared tunnel URL
VITE_ALLOWED_REDIRECTS	<code>http://localhost:5173,http://localhost:3000</code>	same
VITE_STUDY_GUIDER_URL	<code>http://localhost:5173</code>	same
VITE_PAIRPATH_URL	<code>http://localhost:3000</code>	same
VITE_ENABLE_DEV_LOGIN	<code>true</code>	<code>true</code>

VITE_ALLOWED_REDIRECTS is a security control, not a convenience

It is the list of addresses the portal is allowed to hand your login token back to. Without it, anyone could link you to the portal with their own address and collect your token the moment you signed in. If the portal says *"That return address is not allowed"*, add your origin to this list - do not disable the check.

6. Study Guider

Backend (port 8010)

```
cd Study-Guider/backend

python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
copy .env.example .env

.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8010
```

File to edit: Study-Guider/backend/.env

Variable	Mode A (local)	Mode B (Cloudflare)	Required?
CODE_COACH_URL	<code>http://127.0.0.1:8000</code>	the shared tunnel URL	Yes
CORS_ORIGINS	leave the default	leave the default	Yes

Variable	Mode A (local)	Mode B (Cloudflare)	Required?
NEO4J_URI	your Neo4j Aura URI	same	Only to save quiz progress
NEO4J_PASSWORD	your Neo4j password	same	Only to save quiz progress
OPENROUTER_API_KEY	your OpenRouter key	same	Only for AI lessons

Login and remediation work **without** Neo4j and without an OpenRouter key. You will see "database" : "Disconnected" on /api/health, the Analytics tab will say "No Data Points Found", and lessons will use fallback text instead of AI-generated content. That is fine for building UI. Ask the team for the real values when you need them.

If the backend crashes instantly with UnicodeEncodeError

This is a known bug: `app/services/ml_service.py` prints emoji, which the default Windows console cannot encode. Work around it by setting `PYTHONIOENCODING=utf-8` before running (PowerShell: `$env:PYTHONIOENCODING="utf-8"`).

Frontend (port 5173)

```
cd Study-Guider/frontend

npm install
copy .env.example .env

npm run dev                # http://localhost:5173
```

File to edit: `Study-Guider/frontend/.env`

Variable	Mode A (local)	Mode B (Cloudflare)
VITE_API_BASE_URL	<code>http://127.0.0.1:8010</code>	<code>http://127.0.0.1:8010</code> (still local)
VITE_CODE_COACH_URL	<code>http://127.0.0.1:8000</code>	the shared tunnel URL
VITE_PORTAL_URL	<code>http://localhost:4200</code>	<code>http://localhost:4200</code> (run your own)
VITE_ENABLE_DEV_LOGIN	<code>true</code> = own login form; <code>false</code> = use the portal	same

VITE_ENABLE_DEV_LOGIN decides which login you see

Set to **true**, Study Guider shows its own small login form and never touches the portal - faster while building. Set to **false** (or delete the line), it redirects to the portal, which is the real production flow. If you are testing the portal handoff and nothing happens, this is why. Vite only reads `.env` at startup, so restart after changing it.

7. PairPath

PairPath needs a PostgreSQL database. The team uses a free **Neon** cloud database - no local install, and it survives reboots. Ask for the connection strings.

API (port 3001)

```
cd Pair_Path/api

npm install
copy .env.example .env          # then fill in the values below

npx prisma generate
npx prisma migrate dev          # creates the tables, first time only

npm run start:dev               # http://localhost:3001
```

File to edit: Pair_Path/api/.env

Variable	What to put	Required?
DATABASE_URL	Neon DIRECT connection string. Used by migrations only.	Yes
DATABASE_URL_POOLED	Neon POOLED string, plus &pgbouncer=true on the end. Used at runtime.	Yes
JWT_SECRET	Any long random string. Must be the same for the whole team if you share a database.	Yes
CODE_COACH_URL	Mode A: http://127.0.0.1:8000 / Mode B: the tunnel URL	Yes
ML_SERVICE_URL	http://127.0.0.1:8020 (NOT 8000 - that is Code Coach)	Live sessions only
MONGODB_URI	Leave blank. Logs an error and continues.	No
REDIS_URL	Leave blank. Falls back to an in-memory store.	No

Do not change the Prisma versions

prisma and @prisma/client must both stay on **5.22.x**. If npm upgrades them, the API dies at startup with Cannot find module '@prisma/client/runtime/library.js'. Fix: set both back to ^5.22.0 in package.json, delete node_modules/@prisma, then npm install && npx prisma generate.

Frontend (port 3000)

```
cd Pair_Path/frontend

npm install
copy .env.local.example .env.local

npm run dev                # http://localhost:3000
```

File to edit: Pair_Path/frontend/.env.local

Variable	Mode A (local)	Mode B (Cloudflare)
NEXT_PUBLIC_CODE_COACH_URL	http://127.0.0.1:8000	the shared tunnel URL
NEXT_PUBLIC_PORTAL_URL	http://localhost:4200	http://localhost:4200
NEXT_PUBLIC_ENABLE_DEV_LOGIN	true = own login form; unset = use the portal	same

The filename must start with a dot

Next.js reads .env.local. A file named env.local without the leading dot is silently ignored - your settings will look applied and will not be. This has already caught someone on this team.

ml-service (port 8020) - only for live pair sessions

```
cd Pair_Path/ml-service

python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt

.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8020
```

You do not need this for login, the dashboard, or session history. Start it only when you are testing the live collaboration features that call the ML model.

8. Adaptive Gamification Engine

Three parts: an Express API backed by MongoDB, a React frontend, and a Flask service that predicts game difficulty with a Random Forest. It has no accounts of its own - it verifies every request against Code Coach and reads the student's struggle data from there.

Backend API (port 3002)

```
cd adaptive-gamification-engine/backend

npm install
copy .env.example .env

npm start # http://localhost:3002
```

File to edit: adaptive-gamification-engine/backend/.env

Variable	Mode A (local)	Mode B (Cloudflare)	Required?
CODE_COACH_URL	http://127.0.0.1:8000	the shared tunnel URL	Yes
PORT	3002	3002	Yes
MONGODB_URI	mongodb://localhost:27017/code-guru	same	Yes - games and questions
ML_SERVICE_URL	http://127.0.0.1:8030	same	Difficulty prediction only
CORS_ORIGINS	leave the default	leave the default	Yes

There is deliberately no JWT_SECRET here

This service does not issue or verify tokens itself - Code Coach does, via GET /api/v1/auth/me. An earlier version verified signatures locally with a shared secret, which meant signing out of Code Coach did nothing here. If you find code reaching for a signing secret, it is a leftover.

MongoDB is genuinely required for this service - unlike the others, its own data (question bank, game sessions, player profiles) lives there. Without it the API starts but the game routes fail. The dashboard still works, because struggle data comes from Code Coach.

Frontend (port 5174)

```
cd adaptive-gamification-engine/frontend

npm install
copy .env.example .env

npm run dev # http://localhost:5174
```

File to edit: adaptive-gamification-engine/frontend/.env

Variable	Mode A (local)	Mode B (Cloudflare)
VITE_CODE_COACH_ORIGIN	http://127.0.0.1:8000	the shared tunnel URL
VITE_GAMIFICATION_API_URL	http://localhost:3002/api/v1	same
VITE_PORTAL_URL	http://localhost:4200	http://localhost:4200
VITE_ENABLE_DEV_LOGIN	true = own login form; false = use the portal	same

The frontend reaches Code Coach through a Vite proxy rather than calling it directly, so the browser stays same-origin and CORS never applies. VITE_CODE_COACH_ORIGIN is where that proxy forwards to - change that one, not the API path below it.

ml-service (port 8030)

```
cd adaptive-gamification-engine/ml-service

python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt

.\.venv\Scripts\python.exe app.py          # http://localhost:8030
```

Optional. If it is down the backend falls back to a simple rule (many unresolved errors or a low average score means an easier game), so the engine keeps working.

9. Running it day to day

Open a terminal per service and start them in this order. Code Coach must be first - the others verify tokens against it.

Study Guider developers

#	Terminal	Command
1	code-coach/backend	.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8000
2	code-coach/portal	npm run dev
3	Study-Guider/backend	.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8010
4	Study-Guider/frontend	npm run dev

PairPath developers

#	Terminal	Command
1	code-coach/backend	.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8000
2	code-coach/portal	npm run dev
3	Pair_Path/api	npm run start:dev
4	Pair_Path/frontend	npm run dev
5	Pair_Path/ml-service	.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8020

If you set the dev-login flag to true in your frontend, you can skip terminal 2 (the portal) entirely.

Gamification developers

#	Terminal	Command
1	code-coach/backend	<code>.venv\Scripts\python.exe -m uvicorn app.main:app --port 8000</code>
2	code-coach/portal	<code>npm run dev</code>
3	adaptive-gamification-engine/backend	<code>npm start</code>
4	adaptive-gamification-engine/frontend	<code>npm run dev</code>
5	adaptive-gamification-engine/ml-service	<code>.venv\Scripts\python.exe app.py</code>

MongoDB must also be running for the game routes.

Mode B only - the Cloudflare tunnel

One person runs Code Coach and exposes it. Everyone else points their `CODE_COACH_URL` at the printed address.

<code>cloudflared tunnel --url http://localhost:8000</code>	
Who	What they do
Tunnel owner	Runs Code Coach with the Firebase key configured (so data persists and is shared). Adds everyone's browser origins to <code>CORS_ALLOWED_ORIGINS</code> in <code>code-coach/backend/.env</code> .
Everyone else	Puts the tunnel URL in <code>CODE_COACH_URL</code> , <code>VITE_CODE_COACH_URL</code> and <code>NEXT_PUBLIC_CODE_COACH_URL</code> , then restarts their dev servers.

The tunnel URL changes every restart

Cloudflare quick tunnels get a new hostname each time. When the tunnel owner restarts, everyone must update their `.env` files and restart their dev servers. This is why no URL is hardcoded anywhere in the codebase.

10. Getting test data

Study Guider shows nothing until Code Coach has raised a remediation trigger, and that only happens after the same mistake repeats three times. Rather than doing that by hand, run the seed tool:

```
cd code-coach/backend
.\.venv\Scripts\python.exe -m app.dev_tools.seed_student

# Mode B - seed against the shared backend instead:
.\.venv\Scripts\python.exe -m app.dev_tools.seed_student --base-url https://<tunnel-url>
```

It creates a student, generates three real struggles (array indexing, loop boundaries, conditional logic), and prints an access token you can paste into curl or Postman. Sign in as `seed.student@example.com / seed-student-1234`. Running it twice is safe - it reuses the account and creates no duplicates.

11. Check it actually works

#	Do this	You should see
1	Open <code>http://127.0.0.1:8000/health</code>	<code>{"status": "ok"}</code>
2	Open <code>http://localhost:4200</code>	The Code Guru login screen
3	Register a new account there	You land on the hub page

#	Do this	You should see
4	Click through to your service	You arrive already signed in - no second login
5	Check the address bar	No access_token left in the URL
6	Run the seed tool, reload Study Guider	Three struggle cards with real concept names

12. Troubleshooting

Every one of these has actually happened to someone on this team.

Symptom	Cause	Fix
Your service returns 503 on every route	Code Coach is not running.	Start it on port 8000. Everything needs it.
Portal says "That return address is not allowed"	Your origin is missing from the allow-list.	Add it to VITE_ALLOWED_REDIRECTS in code-coach/portal/.env, restart.
Study Guider shows its own login, not the portal	VITE_ENABLE_DEV_LOGIN is true.	Set it to false in Study-Guider/frontend/.env and restart Vite.
PairPath: "PairPath is not responding"	The PairPath API on 3001 is not running.	Start it. Your login is fine - click Try again after.
PairPath .env.local seems to be ignored	The file is named env.local without the dot.	Rename it to .env.local and restart.
Cannot find module '@prisma/client/runtime/library.js'	prisma and @prisma/client are on different majors.	Pin both to ^5.22.0, delete node_modules/@prisma, npm install, npx prisma generate.
UnicodeEncodeError on Study Guider startup	Emoji in print() vs the Windows console encoding.	Set PYTHONIOENCODING=utf-8 before running.
uvicorn.exe: "Fatal error in launcher"	The venv was created before the folder moved. Venvs are not relocatable.	Delete .venv, recreate it, reinstall requirements. Or use python -m uvicorn.
VS Code extension: no yellow underlines, no error	You are signed out. Auto-analysis fails silently by design.	Ctrl+Shift+P, "Code Coach: Sign In".
ML requests behave strangely in PairPath	ml-service is on 8000, colliding with Code Coach.	Run it on 8020 and set ML_SERVICE_URL to match.
Gamification dashboard is empty for a student who has struggles	No triggers yet, or you are signed in as a different student.	Run the seed tool (section 10) and sign in as that student.
Gamification game routes fail but the dashboard works	MongoDB is down. Only this service truly needs it.	Start MongoDB on 27017.
Gamification API returns 503 on every route	It cannot reach Code Coach to verify your token.	Check CODE_COACH_URL and that Code Coach is running. 503 means unreachable, 401 means the token was refused.

13. Rules of the road

Rule	Why
Work on the dev branch. Never commit to main.	main is the reviewed branch. All three repos follow this.
Never commit a .env file.	They hold database passwords and API keys. Every repo already gitignores them - keep it that way.
Do not edit codeguru-auth.js in your own repo.	It is a copy. The master lives in code-coach/portal/src/lib/. Change it there and run code-coach/sync-codeguru-auth.sh, or the three services drift apart.
Never read the student id from a request body.	It comes from the bearer token. Trusting the body lets anyone read anyone else's data - Study Guider had exactly this bug before integration.
Pull code-coach before you start work.	It owns the login and the API contract, so its changes affect everyone.

Full API reference: code-coach/integration/README.md and API_CONTRACT.md in the same folder, which is generated from the running service and therefore cannot drift from the code.